# SmartInterview: Master Technical Explanation & Academic Review Guide

A Comprehensive Defense Document: Literature Survey, Mathematical Metrics, System Architecture, Image Verification & Presentation Script

**Team ID: Team G-1392** | Department of Computer Science & Engineering (AI & ML)
Keshav Memorial Institute of Technology (KMIT) — Mentored by Dr. TVG Sridevi

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## 1. Executive Overview & Problem Statement

Campus placement drives represent a make-or-break milestone for engineering seniors. However, technical interview preparation suffers from systemic failure modes:
1. **Disconnect from Reality:** University academic curricula do not reflect dynamic industry job descriptions.
2. **Severe Anxiety & Vocal Articulation Gaps:** Static platforms (LeetCode, HackerRank) test raw algorithm execution but fail to teach students how to articulate architectural trade-offs, explain system design constraints, or defend reasoning verbally.
3. **Unscalable Human Mock Interviews:** Institutional cohort sizes (hundreds of students) make human coaching economically prohibitive ($50–$150/hour).
4. **Ungrounded AI Hallucinations:** Naive conversational LLM bots hallucinate incorrect API methods, display latency (> 4s), and grade answers arbitrarily on 'vibes'.

**SmartInterview** overcomes all these challenges by establishing an autonomous, grounded, and deterministic mock interview platform combining local vector RAG, stateful Bloom's Taxonomy cognitive progression, 5-factor mathematical evaluation, and sub-1.2s neural voice interaction.

## 2. Foundational Research Papers: In-Depth Comparison

### Base Paper 1: Nagarajan et al. (Feb 2026, IJMRR)

• **Title:** *Development of an Autonomous AI Interview Engine Using Sentence-BERT and Large Language Models for End-to-End Candidate Evaluation*
• **Focus:** Built for enterprise corporate recruiters and talent acquisition teams.
• **Workflow:** Ingests candidate resume and JD; parses text; computes dense semantic vectors via Sentence-BERT (all-MiniLM-L6-v2); ranks profile match via cosine similarity; runs autonomous voice screening using browser Web Speech API, Cloud LLM, and Cloud TTS; outputs scores to recruiter dashboard.
• **Limitations:** Built as an exclusionary corporate screening filter; provides zero candidate learning roadmaps; lacks coding sandboxes; session state is volatile in memory.

### Base Paper 2: Wahid et al. (March 2026, IJERT)

• **Title:** *AI Mock Interview: An Intelligent Voice-Driven Interview Simulation System using Gemini AI and Whisper*
• **Focus:** Built for university students and job seekers practicing for interviews.
• **Workflow:** Streamlit application + Gemini API; student uploads PDF resume via PyPDF2 and selects 1 of 6 generic domains; captures audio at 16 kHz and transcribes using OpenAI Whisper base model; evaluates responses across technical depth, communication, and problem solving; exports ReportLab PDF report with radar charts.
• **Limitations:** Strictly one-way voice interaction (ASR input only; no TTS audio output—candidates must read questions on screen); lacks custom job description matching (rigid dropdowns); Streamlit runtime session volatility.

### How SmartInterview Advances Beyond Both Base Papers

SmartInterview combines the custom JD semantic matching and two-way voice of Paper 1 with the student-centric diagnostic reporting and Whisper accent-robustness of Paper 2, while eliminating the weaknesses of both through local ChromaDB RAG grounding, deterministic Bloom's cognitive scaffolding, and multi-signal objective grading.

## 3. Empirical Benchmark Metrics & Rationale

### A. RAG Retrieval Metrics (33 Technical Domain Queries)

• **Hit@1 = 96.97% (32/33):** Demonstrates that the exact, validated textbook reference chunk is retrieved as the single top match almost 97% of the time.
• **Hit@3 = 100.00% (33/33):** Confirms that in 100% of benchmark queries, the ground-truth technical answer key is contained in the top 3 retrieved chunks.
• **Mean Reciprocal Rank (MRR) = 0.9798:** A near-perfect score (1.0 maximum) confirming that relevant technical facts appear right at the top of the vector search results.
• **Retrieval Latency = 24.6 ms:** Sub-100ms vector search executed entirely locally via ChromaDB HNSW indexing.
• *Why these metrics?* Unconstrained LLMs invent non-existent APIs or incorrect time complexities. By measuring Hit@1, Hit@3, and MRR, we mathematically prove that every technical question and answer key is anchored in proven academic literature, eliminating hallucinations.

### B. Objective 5-Signal Scoring Rubric (Replacing 'Vibe Grading')

SmartInterview evaluates candidate answers using a strict weighted mathematical formula:
Score = 0.30 * Technical + 0.20 * Completeness + 0.20 * Relevance + 0.15 * SBERT_Cosine + 0.15 * Concept_Coverage
• **30% Technical Correctness:** Deep LLM logical reasoning and factual accuracy.
• **20% Completeness:** Adherence to question constraints and addressing all sub-problems.
• **20% Relevance:** Conciseness, directness, and lack of conversational filler.
• **15% SBERT Cosine Similarity:** Sentence-BERT embedding similarity against the verified textbook answer key.
• **15% Explicit Concept Coverage:** Deterministic extraction confirming key technical entities and keywords are articulated.
• *Why this split?* LLMs alone grade on 'vibes'—giving divergent scores to identical answers across runs. By blending semantic vector similarity with deterministic keyword coverage and targeted rubric constraints, scoring is reproducible, transparent, and fair.

### C. Cognitive Scaffolding & Voice Latency Thresholds

• **Bloom's Adaptive Progression:** Score >= 80% (Mastery) advances Bloom level (+1); 50%–79% (Adequate) maintains level; Score < 50% (Remediation) regresses Bloom level (-1).
• **Voice Latency <= 1.2s:** In-memory io.BytesIO streaming with Microsoft Edge-TTS eliminates disk I/O bottlenecks.
• **End-to-End Turnaround < 2.5s:** Groq Cloud LPU delivers a Time-to-First-Token (TTFT) of ~320ms, preventing awkward pauses during live mock interviews.

## 4. System Diagram & Image Verification (Latest Commit on Main)

Audit conducted against GitHub main branch latest commit (b298c44db6326a42af5770ee354cbe1b664c6d0b):

| Diagram / Asset Name | File Path | Dimensions & Size | Label & Deliverables Reference | Verification Status |
|---|---|---|---|---|
| System Architecture | 05_System_Diagrams/Architecture.png | 840x523 \| 32.1 KB | SRS Fig 2.1, PPT Slide 4, Paper Fig 1 | VERIFIED (100% Coherent) |
| Entity Relationship (ERD) | 05_System_Diagrams/ERD.png | 462x1101 \| 44.2 KB | SRS Fig 2.2, Encyclopedia Part 5 | VERIFIED (100% Coherent) |
| Interview Execution Workflow | 05_System_Diagrams/Workflow.png | 980x809 \| 60.1 KB | SRS Fig 2.3, PPT Slide 5, Paper Fig 2 | VERIFIED (100% Coherent) |
| Frontend Hero Banner | frontend/src/assets/hero.png | 343x361 \| 13.0 KB | Landing Page UI Illustration | VERIFIED (100% Coherent) |
| Favicon Icon | frontend/public/favicon.svg | Vector SVG \| 9.5 KB | Browser Tab Navigation Icon | VERIFIED (100% Coherent) |

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|---|---|---|---|---|
| System Icons Sprite | frontend/public/icons.svg | Vector SVG \| 5.0 KB | Action Icons (Mic, Volume, Stats) | VERIFIED (100% Coherent) |

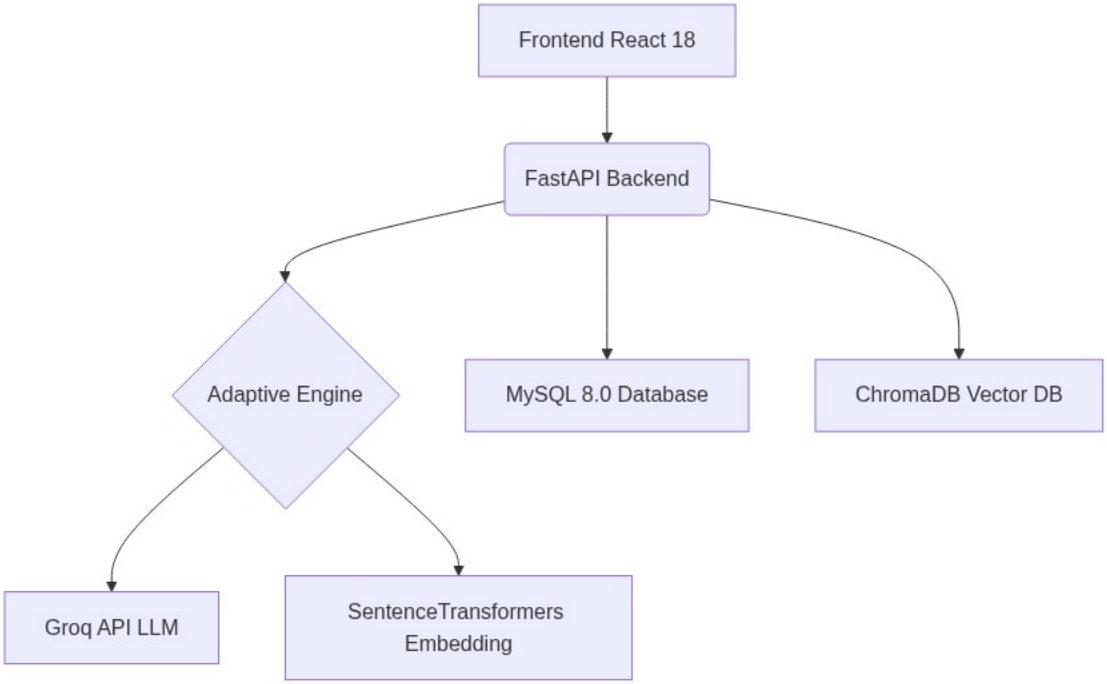


***Figure: Decoupled 4-Tier Distributed Architecture of SmartInterview***

## 5. Spoken-Word Presentation Script for Academic Defense

Good morning, respected members of the panel. When engineering students prepare for campus placements, technical interviews are consistently the highest hurdle they face. While mock interviews are proven to reduce anxiety and build confidence, human mock interviews are expensive to scale, and static question banks simply cannot simulate a real, dynamic conversation. Today, I will be presenting a comparative review of two foundational 2026 research papers in this domain, and showing how our project, SmartInterview, directly overcomes their architectural limitations to build a grounded, adaptive, and deterministic interview coach.

Let us begin with Paper 1, published by Nagarajan and colleagues in February 2026. Their work is built primarily for enterprise recruiters and HR talent acquisition teams looking to automate candidate screening. Their workflow begins with the candidate's PDF resume and a specific job description. The system parses both documents and uses Sentence-BERT—specifically the all-MiniLM-L6-v2 model—to generate dense semantic vectors. By calculating the cosine similarity between the resume and the job description, it automatically measures how well an applicant's profile matches the job requirements. A Flask backend then drives an autonomous, two-way voice interview using the browser's Web Speech API for speech-to-text, a cloud LLM to generate adaptive questions, and cloud text-to-speech to speak to the candidate. When the interview ends, the LLM evaluates the transcript across technical proficiency, communication, and project depth, presenting the results on a recruiter web dashboard. While this is an impressive corporate pipeline, its limitations stem directly from its recruiter-first design. It functions as an exclusionary gatekeeper—designed to filter people out rather than help students improve. As a result, it provides no offline diagnostic reports or upskilling roadmaps for candidates. It lacks any live code execution sandbox to verify practical claims, and its session state is held in volatile memory, meaning any network hiccup drops the session entirely.

Turning to Paper 2, published by Wahid and colleagues in March 2026, we see an approach centered on student preparation. Here, the candidate uploads a PDF resume using PyPDF2 and chooses one of six broad technical domains, such as Machine Learning or Cloud Computing. Built on Streamlit and powered by the Gemini API, the platform runs an adaptive loop of eight to twelve questions. For voice capture, it records audio at 16 kilohertz using SoundDevice and

transcribes it using OpenAI’s Whisper base model. Whisper is a strong choice here because it handles diverse accents and technical jargon with exceptional accuracy. At the end of the session, Gemini generates a structured JSON evaluation scoring overall performance, technical depth, communication, and problem-solving. This feeds into ReportLab, which auto-generates a clean, downloadable multi-page PDF report complete with skill radar charts and tailored learning roadmaps. However, Paper 2 has two major flaws. First, the voice experience is strictly one-way. Whisper transcribes the candidate's speech, but the AI never speaks back—the student is forced to read questions off the screen, which breaks the realism of a spoken interview. Second, it lacks custom job description matching; students are restricted to rigid domain dropdowns, meaning they cannot tailor their practice to a specific company or job posting.

When we place both papers side-by-side, we see a stark trade-off: Paper 1 offers custom job description matching and full two-way voice, but locks all feedback inside an HR dashboard designed to screen candidates out. Paper 2 puts the student first with accent-robust Whisper input and a downloadable PDF roadmap, but it removes audio output entirely and forces candidates into generic domain dropdowns. Moreover, both papers share critical vulnerabilities: they rely on volatile in-memory runtimes, their LLMs are prone to unverified hallucinations, and their scoring depends on single-prompt black-box LLM evaluation—what we call 'vibe grading.'

This brings us to SmartInterview. We engineered our platform to bridge this exact divide by combining student-centered pedagogy with deterministic, low-latency AI. First, we attacked the problem of AI hallucinations at its foundation using Retrieval-Augmented Generation, or RAG, backed by ChromaDB over a curated knowledge base of 402 computer science topic summaries across eight domains. Our retrieval pipeline achieved a Hit at 1 of 96.97%, a Hit at 3 of 100%, and an MRR of 0.9798. In plain English, Hit at 1 means that nearly 97% of the time, the exact, verified textbook reference needed for a question is retrieved as the single top match. A Hit at 3 of 100% means the ground truth is guaranteed to be in the top three results. And an MRR of nearly 0.98 mathematically proves that relevant context consistently appears at the very top of the list.

Second, we eliminated black-box 'vibe grading.' We replaced subjective marks with an objective, five-part scoring rubric: 30% Technical Correctness, 20% Completeness, 20% Relevance, 15% SBERT Cosine Similarity against our ground-truth reference, and 15% Explicit Concept Coverage. We chose this split because semantic similarity ensures the candidate explained the underlying concept correctly, while concept coverage verifies that critical technical keywords and mechanisms were explicitly articulated. Third, we resolved unstructured topic-hopping by implementing a stateful cognitive engine powered by Bloom’s Revised Taxonomy. Real interviewers don’t jump erratically between trivia and architectural design; they probe deeper. Our system scaffolds dynamically—starting from foundational recall and advancing up through Apply, Analyze, and Create when a candidate demonstrates mastery with scores of 80% or higher, while stepping back to reinforce fundamentals if a score dips below 50%. Fourth, to deliver true two-way conversational immersion without awkward pauses, we engineered an in-memory neural voice synthesis pipeline using io.BytesIO, achieving a voice latency of 1.2 seconds or less, and total end-to-end turnaround under 2.5 seconds. Finally, SmartInterview provides three targeted operational modes: Job-Specific Mode, Topic-Based Mode, and Semester Syllabus Mode with dynamic collection isolation. In conclusion, while previous research treats the AI interview either as an exclusionary corporate screening filter or a silent web form, SmartInterview transforms it into a grounded, low-latency, and pedagogically sound training ground that empowers students to succeed. Thank you, and I am now ready to take your questions.